

The Colors of Romance:

A Data Science Approach

Report

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Table of Contents

Table of Contents.....	2
Introduction.....	4
Data Collection and Extraction.....	5
Data Collection.....	5
Data Extraction.....	5
Example of CSV.....	6
Broad Color Analysis.....	7
Initial Visualization and Data Preparation.....	7
3D Plots of Colors Used.....	7
Dominant Color per Frame Plotted.....	8
1. Color Palette and Visual Tone:.....	11
2. Consistency and Variation in Color Use:.....	12
3. Role of Scene Detection:.....	12
K-Means Clustering.....	13
Segmentation and Initial Clustering Process.....	13
Classification into Primary Color Groups.....	13
Tone Classification Percentages.....	15
Tone Classification Graphs.....	16
Key Similarities:.....	18
Interpreting the Similarities:.....	19
Segmented Time Series Analysis.....	20
Analysis of Saturation, Brightness, and Contrast.....	22
How the Values Were Calculated:.....	22
Graphs of Saturation, Brightness, and Contrast.....	22
1. Saturation:.....	23
2. Brightness:.....	23
3. Contrast:.....	24
Analysis of Color Temperature.....	24
Temperature Calculation.....	24
Graphs of Color Temperature.....	25
1. Common Trend Towards Neutral Tones:.....	27
2. Distinct Preferences:.....	27
Final K-Means Clustering.....	28
Final K-Means Colors.....	28
1. Distribution of Colors:.....	29
2. Brightness:.....	29
3. Saturation:.....	30
4. Color Temperature:.....	30
Gaussian Mixture Modelling.....	31

Colors Produced by GMM.....	31
1. GMM Clustering for English Films:.....	32
2. GMM Clustering for Hindi Films:.....	32
3. GMM Clustering for Korean Films:.....	33
Linguistic Link To the Colors.....	34
The Videos.....	34
Investigating Etymological and Cultural Links.....	34
English and Korean Film.....	34
Hindi Film.....	35
Conclusion.....	36
Conclusion.....	37
Key Findings.....	37
Final Thoughts.....	38
Citations.....	39
Appendix.....	40
List of English Movies.....	40
List of Hindi Movies.....	41
List of Korean Movies.....	42
English Segmented K-Means Colors.....	43
Hindi Segmented K-Means Colors.....	45
Korean Segmented K-Means Colors.....	47

Introduction

The portrayal of romance in cinema is often deeply intertwined with cultural nuances and artistic choices, particularly in the use of color. Colors not only enhance the visual appeal of a film but also evoke emotions and set the tone for romantic scenes. In this project, titled "The Colors of Romance," I aim to explore how different languages and cultural contexts influence the color palettes used in romance films. Specifically, this study will focus on films in English, Hindi, and Korean—three languages representing some of the largest film industries in the world.

When I think of English romance films, particularly those from Hollywood and British cinema, the imagery that comes to mind often includes dramatic scenes set against dark and moody color schemes. I envision scenes in the rain, bathed in bright spotlights, or the soft, muted tones of English fields, all contributing to a sense of heightened drama and intensity.

In contrast, Indian romance films, particularly those influenced by the works of directors like Yash Chopra, are often characterized by vibrant and bold color schemes. These films frequently feature elaborate dance sequences set against the backdrop of stunning locations like Switzerland, where the over-the-top use of color enhances the fantasy and grandeur of romance.

As for Korean romance films, I am less certain about their typical color schemes, which makes this analysis particularly intriguing. I am eager to see what patterns emerge from the data and how they compare to the more familiar landscapes of English and Hindi cinema.

The choice of these three languages is deliberate, as they not only represent major film industries but are also languages in which I personally enjoy watching films. While this study will focus on these languages, future analyses might extend to other languages, such as Malayalam, Turkish, and Japanese, further broadening the scope of this exploration.

The following sections of this report will detail the methodology used for data collection and analysis, followed by an in-depth examination of the color patterns observed in films across these three languages. Through this analysis, I hope to uncover whether the cultural context embedded in language influences the color palettes chosen to depict love and romance on screen.

Data Collection and Extraction

Data Collection

The first step in this project was to select a representative sample of romance films for each language—English, Hindi, and Korean. For English and Hindi films, the task was relatively straightforward. I utilized IMDb's advanced filtering feature, which allowed me to set specific criteria: films released from 1990 to the present, in the specified language, with at least 50,000 ratings, and classified under the romance genre. This filtering process yielded a list of 25 films for each language, which I considered a good sample for analyzing color usage.

Selecting Korean films proved more challenging due to the lack of comprehensive resources in English. The IMDb filters were less effective, as many Korean films had fewer than 3,000 reviews, making them less representative of popular cinema. After exploring several sources, I ultimately compiled my list from a user-generated list with over 400,000 views, supplemented by a Marie Claire article. While subjective, this approach provided a reasonably representative sample of Korean romance films. A full list of the selected films and the sources used can be found in the appendix.

After finalizing the lists, I acquired digital copies of each film. Depending on the film's release date, this process ranged from straightforward to challenging, but I successfully obtained all necessary copies for the analysis.

Data Extraction

With the digital copies in hand, the next step was to extract the color data from each film. To achieve this, I employed two primary scripts: `color_extraction_script_2` and `remaining_frames_extraction`. These scripts were designed to extract dominant colors from selected frames of the films efficiently and accurately.

The `color_extraction_script_2` script utilized the ColorThief library in combination with FFmpeg for scene detection. Scene detection was crucial for optimizing the extraction process by limiting the number of frames analyzed. The script employed FFmpeg's `select='gt(scene, {scene_threshold})'` filter, which allowed it to extract frames only when a significant change in the scene was detected. This approach ensured that only frames with substantial visual differences were selected, reducing redundancy and focusing on key moments that might reveal different color palettes.

Once the relevant frames were identified, the script used ColorThief to extract the nine most dominant colors from each frame. These colors were then stored in a CSV file, with each

row corresponding to a frame and containing the RGB values of the dominant colors. To conserve disk space, the script deleted each frame after processing. The process was further optimized using Python's `multiprocessing` module, allowing for parallel processing of frames and significantly speeding up the extraction.

However, in some cases, a few frames were left unprocessed after the initial run of `color_extraction_script_2`. To address this, I developed the `remaining_frames_extraction` script. This script checked each movie's frame directory for any leftover frames that had not been processed and repeated the color extraction process for these frames. The extracted colors were then appended to the existing CSV files, ensuring that all relevant frames were accounted for.

The combination of these two scripts resulted in a comprehensive dataset of dominant colors extracted from the selected films. In total, 75 CSV files were generated—25 for each language—each containing the dominant colors from the most visually significant frames of the corresponding films. This dataset serves as the foundation for the subsequent analysis, allowing for a detailed examination of the color palettes used in romance films across different languages.

Example of CSV

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	frame_path	color_1_r	color_1_g	color_1_b	color_2_r	color_2_g	color_2_b	color_3_r	color_3_g	color_3_b	color_4_r	color_4_g	color_4_b
2	output_0000001.png	94	76	59	210	190	185	28	24	21	146	115	92
3	output_0000002.png	38	36	28	191	177	178	129	114	101	94	83	68
4	output_0000003.png	126	109	90	28	24	19	196	180	163	168	154	138
5	output_0000004.png	41	32	30	94	81	72	105	100	90	121	103	93
6	output_0000005.png	32	27	26	196	174	161	121	93	80	86	71	62
7	output_0000006.png	204	198	188	40	32	28	134	119	107	106	110	106
8	output_0000007.png	202	184	174	55	44	37	155	133	122	117	115	112
9	output_0000008.png	32	33	26	140	117	102	194	176	163	100	94	65
10	output_0000009.png	186	173	161	39	32	27	144	132	122	98	87	73

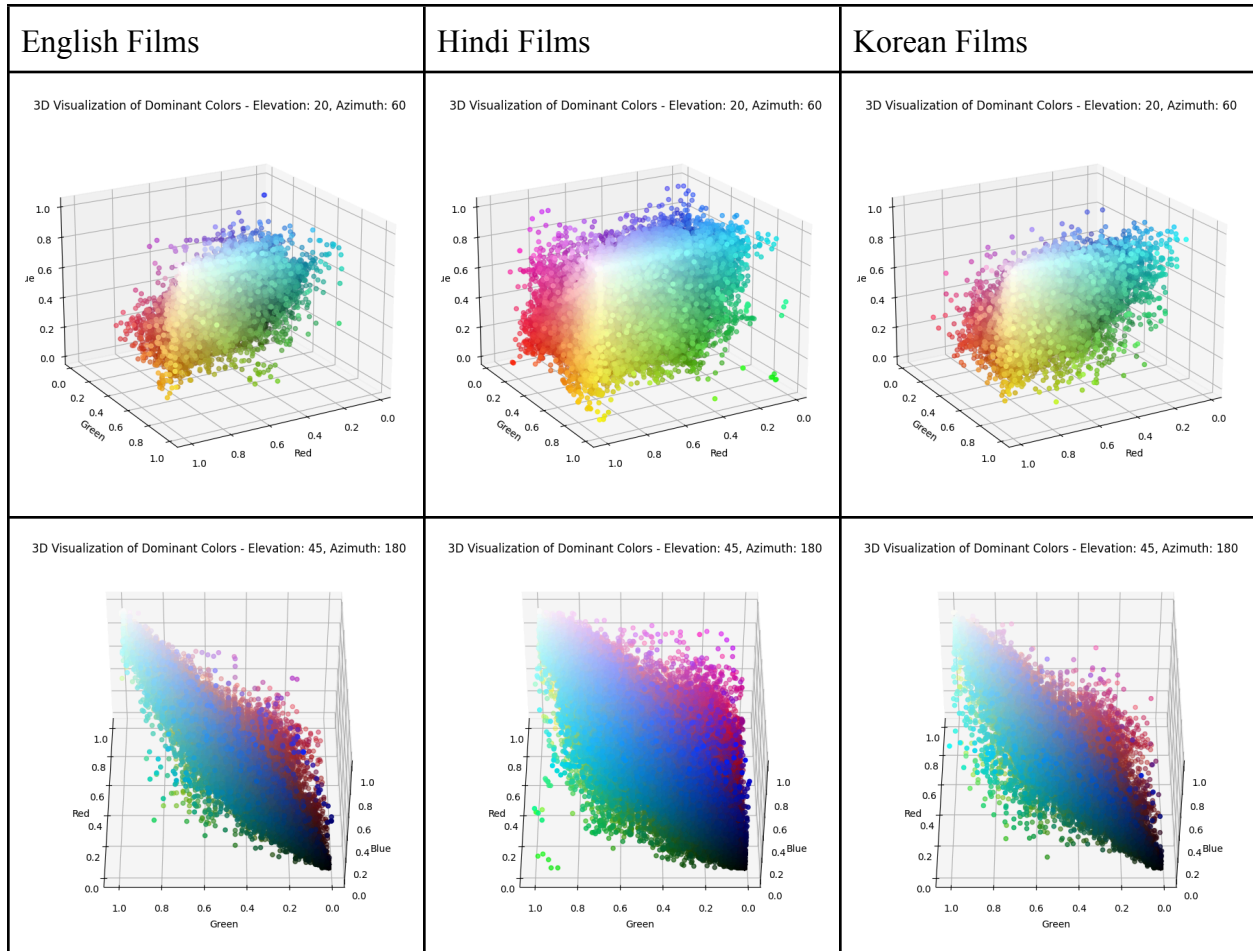
Broad Color Analysis

In this section, we aim to explore the dominant color patterns present in English, Hindi, and Korean romance films through a series of visual and clustering analyses. By examining the saturation, brightness, contrast, and color temperature of the colors used, we seek to understand how different cultures and languages may influence the portrayal of romance on screen.

Initial Visualization and Data Preparation

For each language, the color data was extracted from the 25 romance films, stored in individual CSV files for each movie, and then combined into a single dataset. Each row in this dataset corresponded to a set of dominant colors extracted from a movie frame, with an added movie label for easier identification during the analysis. These initial 3D visualizations provided a broad understanding of the color distribution across the films in each language.

3D Plots of Colors Used



The 3D plots above illustrate the distribution of dominant colors across English, Hindi, and Korean romance films. Each point in the plots represents a specific color in RGB space, offering a comprehensive visual summary of the color palettes utilized in these films.

English Films: In the English films plot, the color distribution appears to be somewhat concentrated, with a noticeable cluster of colors in the middle of the RGB space. This suggests that English romance films might have a more restrained or uniform color palette, potentially aligning with the hypothesis that these films often utilize darker and more dramatic tones. The concentration in the middle could indicate a preference for more neutral colors, possibly leaning towards cooler tones often associated with the somber, dramatic mood characteristic of many English-language romance films.

Hindi Films: The Hindi films plot reveals a much broader spread of colors across the RGB spectrum. The wider dispersion suggests a more varied and vibrant use of color, which is consistent with the colorful and dynamic visual style typical of Indian cinema. The presence of more points in the outer areas of the RGB space indicates the use of both highly saturated colors and a wide range of hues, supporting the hypothesis that Hindi romance films often feature bold and bright color schemes, particularly in dance sequences and scenes set in vivid, picturesque locations.

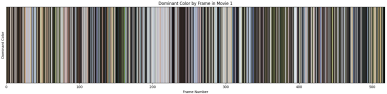
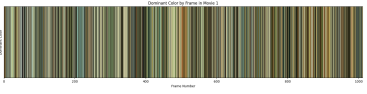
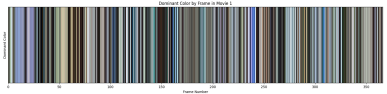
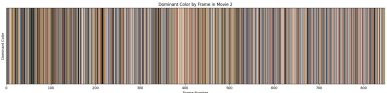
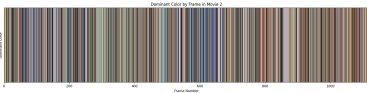
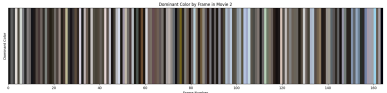
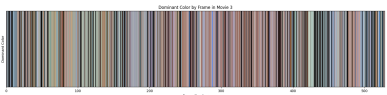
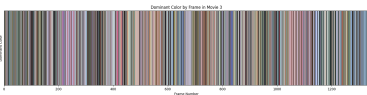
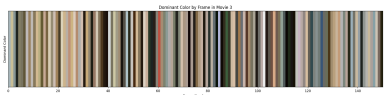
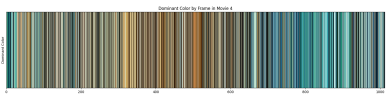
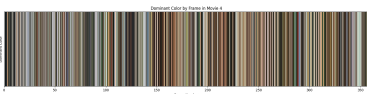
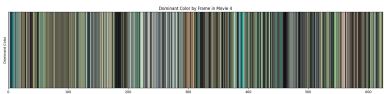
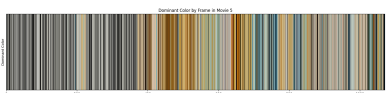
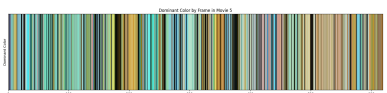
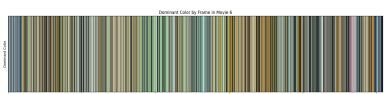
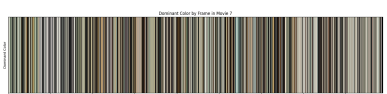
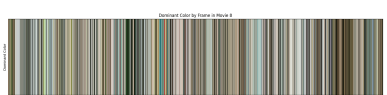
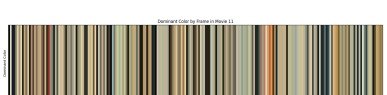
Korean Films: The plot for Korean films shares similarities with the English films, showing a more concentrated cluster of colors. However, there are subtle differences in the distribution, suggesting a potential preference for slightly cooler tones compared to English films. This might reflect the unique visual aesthetic often found in Korean romance films, which tend to balance realism with a soft, emotionally resonant color palette. The concentration around certain areas of the RGB space hints at a more deliberate use of specific colors to evoke particular moods or themes, though it does not exhibit the same level of color diversity seen in Hindi films.

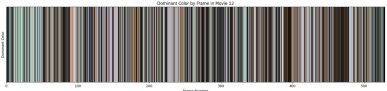
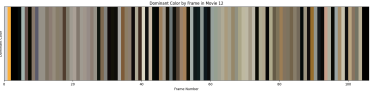
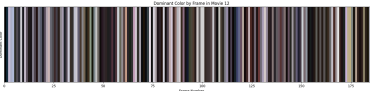
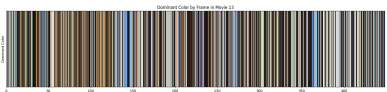
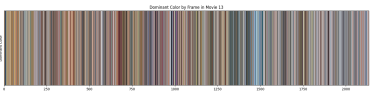
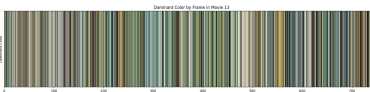
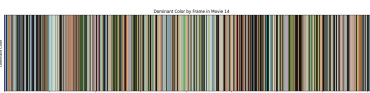
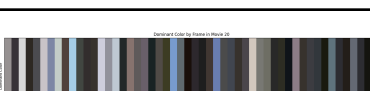
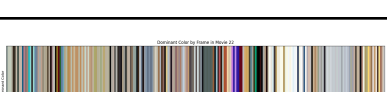
These visualizations provide a foundational understanding of the color palettes used in romance films across these three languages, setting the stage for more detailed clustering and analysis of specific color attributes like saturation, brightness, contrast, and color temperature.

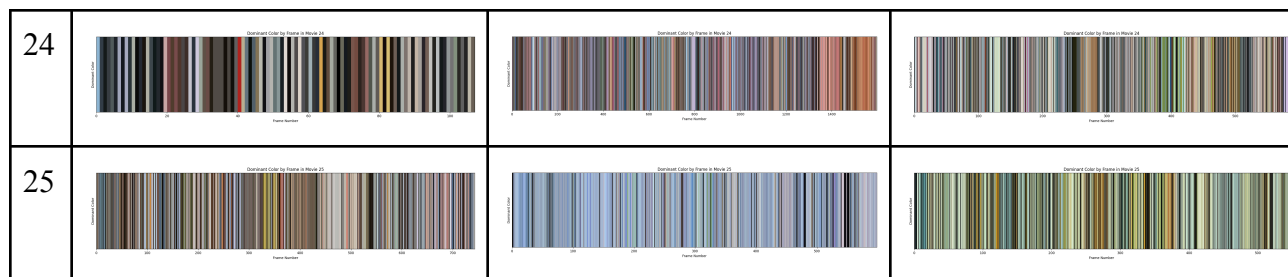
Dominant Color per Frame Plotted

In this section, I aim to explore the general color palettes of each film to see if discernible patterns emerge by eye. Plotting the most dominant color per frame gives me a general idea of the colors used throughout each film. This approach allows for preliminary analyses that might reveal visual trends or distinctive color usage that are characteristic of romance films in English, Hindi, and Korean cinema.

Note: The graphs here are condensed to save space, feel free to see the larger graphs on the associated Jupyter Notebooks - **(Language)_K_means_&_Color_Analysis.ipynb**

	English Films	Hindi Films	Korean Films
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			

12	 Download Color by Frame in Movie 12	 Download Color by Frame in Movie 12	 Download Color by Frame in Movie 12
13	 Download Color by Frame in Movie 13	 Download Color by Frame in Movie 13	 Download Color by Frame in Movie 13
14	 Download Color by Frame in Movie 14	 Download Color by Frame in Movie 14	 Download Color by Frame in Movie 14
15	 Download Color by Frame in Movie 15	 Download Color by Frame in Movie 15	 Download Color by Frame in Movie 15
16	 Download Color by Frame in Movie 16	 Download Color by Frame in Movie 16	 Download Color by Frame in Movie 16
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22	 Download Color by Frame in Movie 22	 Download Color by Frame in Movie 22	 Download Color by Frame in Movie 22
23	 Download Color by Frame in Movie 23	 Download Color by Frame in Movie 23	 Download Color by Frame in Movie 23



The analysis of the dominant color per frame across English, Hindi, and Korean romance films reveals distinct visual styles that are influenced by cultural, narrative, and cinematic conventions within each language group. Here's a comparative overview of the key findings:

1. Color Palette and Visual Tone:

- English Films:** Scanning the graphs, I notice more often than not that the color palette in English films is more muted and neutral, with a strong emphasis on earth tones, grays, and subdued shades. This could reflect a visual tone that is often realistic and naturalistic, aiming to ground the narrative in a believable environment. The use of darker colors contributes to a mood that could be interpreted as introspective and often serious.
- Hindi Films:** In contrast, Hindi films display a much broader and more vibrant color palette. The frequent use of bright reds, blues, greens, and other saturated colors reflects the dynamic and expressive nature of Bollywood cinema. The bold use of color enhances the emotional intensity and visual spectacle of the films, often amplifying the romantic and dramatic elements of the story. This aligns with the cultural preference in Indian cinema for vibrant, larger-than-life portrayals of romance and emotion.
- Korean Films:** The color palette in Korean films tends to be more subdued and cooler, with a noticeable preference for blues, grays, and greens. Similar to English films, there seems to be a focus on mood and atmosphere. In my experience, the use of color in Korean films often supports the narrative's introspective and sometimes melancholic tone, reflecting the themes of quiet passion and the complexities of relationships common in Korean cinema.

2. Consistency and Variation in Color Use:

- **English Films:** The English films tend to show a high level of consistency in color use, with gradual shifts between similar tones. This suggests a restrained and deliberate approach to color, where changes in the visual palette are used sparingly to highlight key moments in the narrative.
- **Hindi Films:** Hindi films, on the other hand, exhibit frequent and dynamic shifts in color, reflecting the energetic and often fast-paced nature of Bollywood storytelling. The varied use of color aligns with the cultural and cinematic conventions of Indian cinema, where visual vibrancy is integral to the overall viewing experience.
- **Korean Films:** Korean films show consistency similar to English films, but with even more subtle transitions in color. The restrained use of color shifts reflects a focus on maintaining a specific emotional tone throughout the film, with color serving as a background element that subtly enhances the narrative.

3. Role of Scene Detection:

- The scene detection algorithm used in this analysis highlights how different films handle visual transitions. English and Korean films, with their more consistent use of color, have fewer frames extracted, reflecting a slower pace and more gradual shifts in visual tone.
- Hindi films, with their frequent color changes and dynamic scenes, have more frames analyzed, reflecting the rapid shifts in mood and setting typical of Bollywood cinema.

The comparative analysis of color usage in English, Hindi, and Korean romance films begins to reveal how cultural and cinematic conventions shape the visual portrayal of romance. While English and Korean films favor a more subdued and consistent use of color to support their narrative tone, Hindi films embrace a vibrant and dynamic palette that enhances the emotional and visual spectacle of the story. These differences highlight the diverse ways in which color is used to evoke emotions and set the tone in romance films across different cultures.

K-Means Clustering

To further analyze the color patterns in each film, I conducted two rounds of K-Means clustering. The first round was to identify the predominant colors within different segments of each movie, enabling a more detailed understanding of the visual themes present across the three languages. Once that was complete, I ran another round on the colors from the first round to get a representative set of colors for each language.

Segmentation and Initial Clustering Process

For each film, I divided the frames into five segments, ensuring a consistent approach to analyzing the colors throughout the movie. The number of frames was divided by five, and K-Means clustering was applied to each segment using 15 clusters. This choice of cluster number was informed by previous work, such as the analysis I conducted on the Ghibli project, where I observed that fewer clusters tend to produce more muted color clusters. By selecting 15 clusters, I aimed to strike a balance, capturing both vibrant and muted tones within each segment.

However, certain segments contained very few frames, which posed a challenge for clustering. To address this, I implemented a minimum threshold: if a segment contained fewer than 5 frames, or fewer than 50 colors, I skipped clustering for that segment.

This initial clustering yielded approximately 2000 colors to represent each language. The full set of colors can be viewed in the respective notebooks for each language:

(Language)_K_means&_Color_Analysis.ipynb.

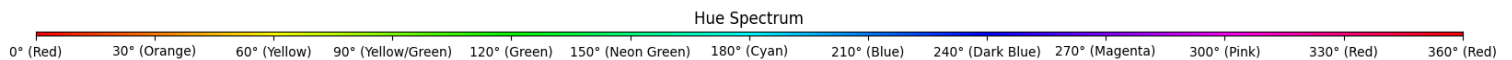
Classification into Primary Color Groups

Following the initial clustering, I aimed to classify these colors into primary color groups to determine whether any of the languages exhibited a preference for certain color tones. To achieve this, I categorized each color into a tone category (e.g., blue, red, yellow) based on its hue, using the HSV (Hue, Saturation, Value) color space.

- **Hue** represents the type of color (e.g., red, blue, green) and is measured as an angle on a color wheel from 0° to 360° . For example:
 - 0° corresponds to red.
 - 120° corresponds to green.
 - 240° corresponds to blue.
- **Saturation** indicates the intensity of the color, ranging from dull/gray to pure color.
- **Value** represents the brightness of the color.

To provide more variation in the classification, I used the following hue segments to categorize the colors:

- 0° - Red
- 30° - Orange
- 60° - Yellow
- 90° - Yellow/Green
- 120° - Green
- 150° - Neon Green
- 180° - Cyan
- 210° - Blue
- 240° - Dark Blue
- 270° - Magenta
- 300° - Pink
- 330° - Red
- 360° - Red



Note: This approach does not capture all the colors present, as it omits shades of black, white, gray, browns, etc. However, it allowed me to generally group colors together. I found that adding more color categories tended to drastically change the distribution each time, increasing the analysis's complexity. I decided to stick with the HSV color space to simplify the process, although I recognize it may not be the perfect solution.

Once classified, I counted the number of colors in each category and calculated the percentage of each color tone relative to the total number of colors. This process allowed for a detailed comparison of color usage across the languages, highlighting any potential biases or preferences for certain color tones.

Tone Classification Percentages

English Film (Total Colors - 2175)

Summary of tone counts and percentages (sorted by percentage):

Red: 692 colors (31.82%)

Orange: 526 colors (24.18%)

Cyan: 262 colors (12.05%)

Blue: 196 colors (9.01%)

Yellow: 180 colors (8.28%)

Neon green: 118 colors (5.43%)

Yellow/green: 73 colors (3.36%)

Green: 56 colors (2.57%)

Dark blue: 43 colors (1.98%)

Pink: 17 colors (0.78%)

Magenta: 12 colors (0.55%)

Hindi Film (Total Colors - 2160)

Summary of tone counts and percentages (sorted by percentage):

Red: 711 colors (32.92%)

Orange: 415 colors (19.21%)

Cyan: 324 colors (15.00%)

Blue: 214 colors (9.91%)

Yellow: 123 colors (5.69%)

Neon green: 99 colors (4.58%)

Yellow/green: 96 colors (4.44%)

Green: 74 colors (3.43%)

Dark blue: 45 colors (2.08%)

Magenta: 32 colors (1.48%)

Pink: 27 colors (1.25%)

Korean Films (Total Colors - 2175)

Summary of tone counts and percentages (sorted by percentage):

Orange: 547 colors (25.15%)

Red: 526 colors (24.18%)

Cyan: 289 colors (13.29%)

Neon green: 193 colors (8.87%)

Yellow: 192 colors (8.83%)

Blue: 141 colors (6.48%)

Green: 109 colors (5.01%)

Yellow/green: 108 colors (4.97%)

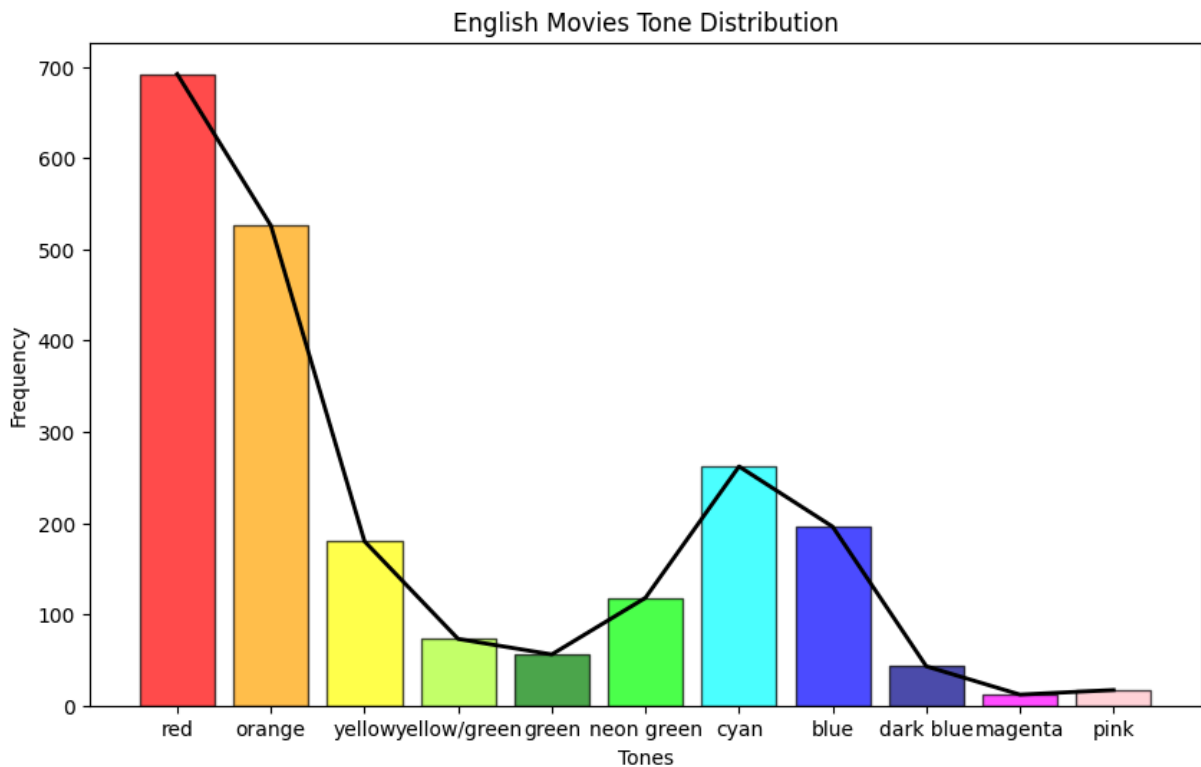
Dark blue: 34 colors (1.56%)

Magenta: 24 colors (1.10%)

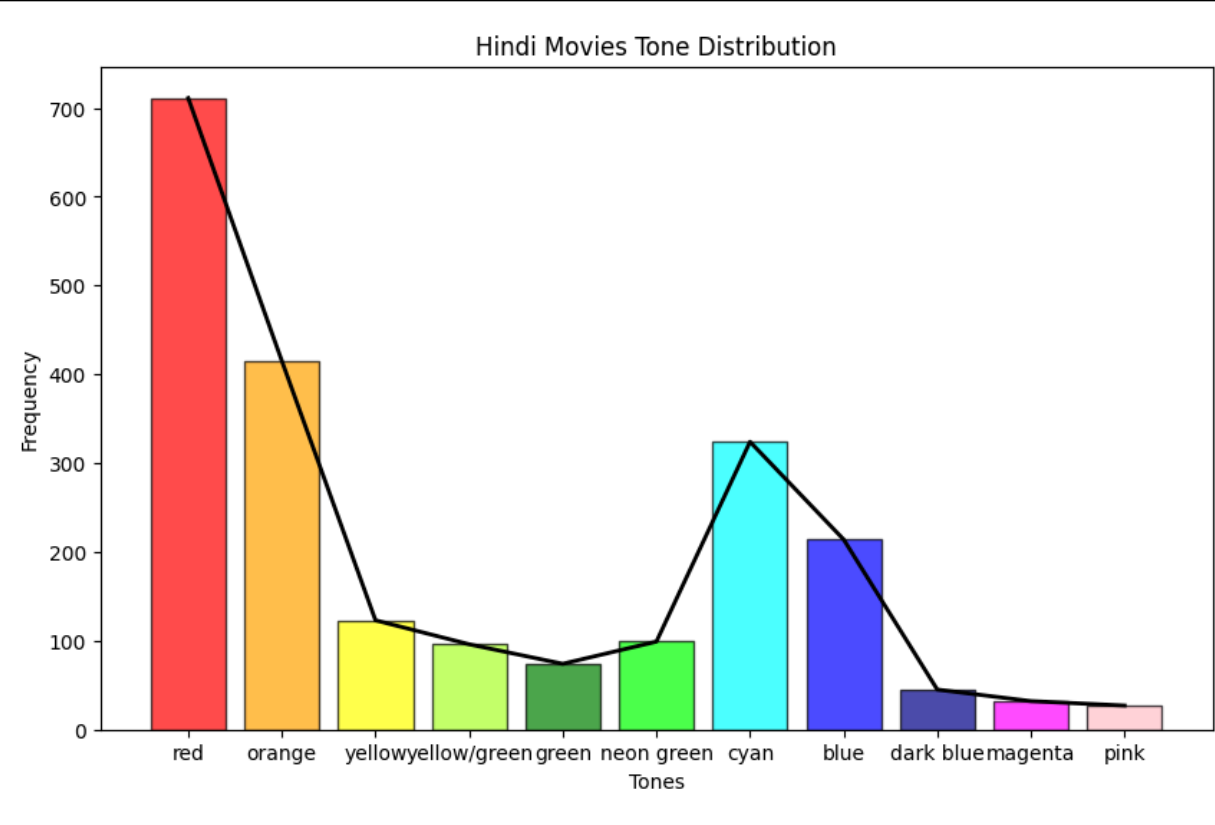
Pink: 12 colors (0.55%)

Tone Classification Graphs

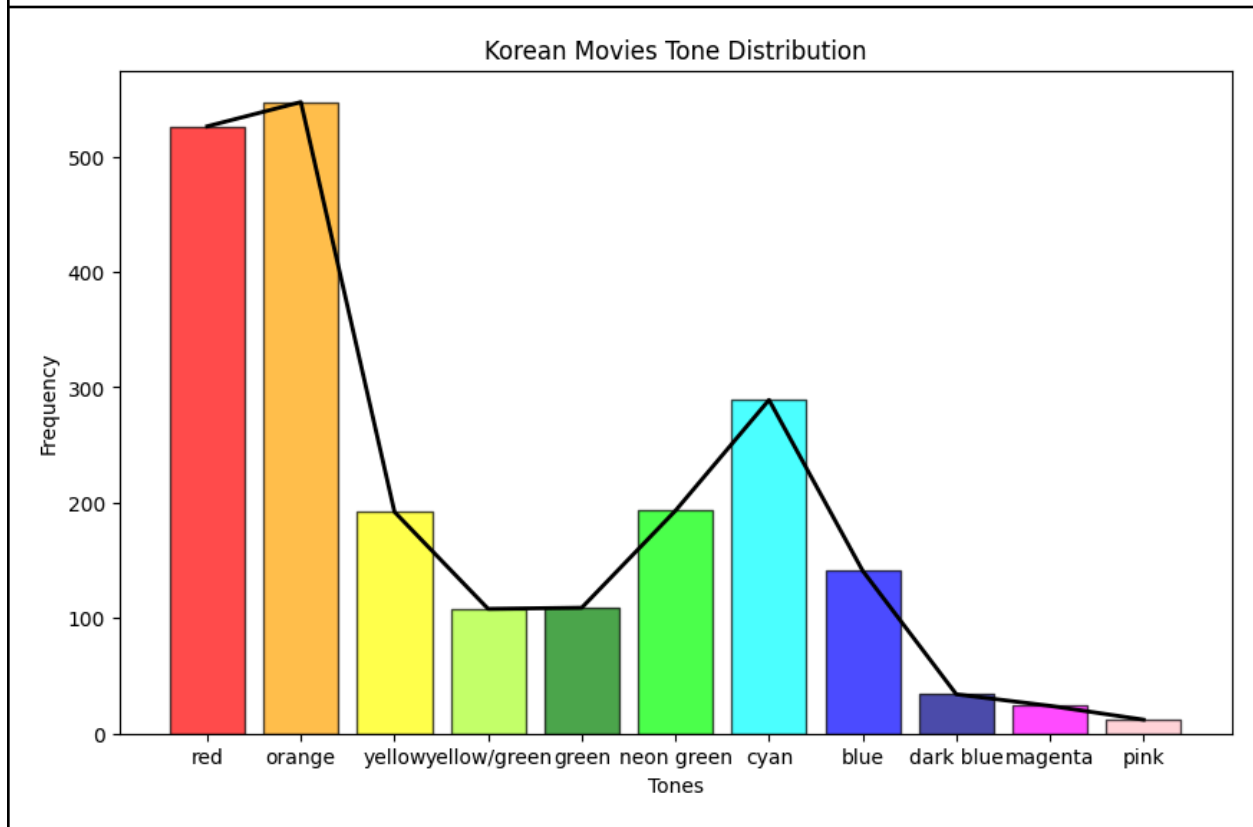
English Films



Hindi Films



Korean Films



The K-Means clustering results reveal that the color distributions across English, Hindi, and Korean romance films are remarkably similar. Warm tones, particularly red and orange, dominate the color palettes in all three languages, accounting for a significant portion of the total colors. Cool tones like cyan and blue also appear with notable frequency, though to a lesser extent.

Key Similarities:

1. Dominance of Warm Tones:

- Across all three languages, red and orange tones are the most dominant, making up nearly 50% of the color palette in each case. This suggests a shared visual language in romance films, where warm colors are used universally to convey emotions like passion, love, and warmth.

2. Presence of Cool Tones:

- Cyan and blue are present in all three languages, with percentages ranging between 9% to 15%. These colors likely serve to balance the warm tones, adding contrast and depth to the visual narrative.

3. Varied but Limited Use of Other Colors:

- The remaining colors (like green, yellow, neon green, etc.) are used more sparingly across all films, contributing less to the overall color scheme. This similarity suggests that while films in different languages might have unique cultural elements, they often rely on a similar core palette for visual storytelling in romance.

Interpreting the Similarities:

- Given the close resemblance in color distributions, it's important to recognize that these similarities might stem from shared cinematic techniques and the universal nature of color symbolism in romance films, rather than distinct cultural or artistic differences. The use of warm tones to evoke passion and cool tones to create contrast is a common strategy across different film industries, suggesting a convergent approach to visual storytelling in this genre.
- While there may be subtle differences in how these colors are used or emphasized within individual films, the overall patterns indicate that romance as a genre has a relatively consistent visual language, regardless of the cultural context.

The K-Means clustering results indicate that the color palettes of English, Hindi, and Korean romance films are more alike than different. This uniformity suggests that certain color choices, particularly the dominance of warm tones, are universally effective in conveying the emotions and themes central to romance films. While individual films may exhibit unique visual styles, the general approach to color in the genre appears to be consistent across different languages and cultures.

What strikes me is that film industries tend to pull inspiration from each other quite often. This exchange of ideas and stylistic choices could be contributing to the similarities we observe in the color palettes across these films. It raises an intriguing question: would the colors used in these films be different if the film industries were completely isolated from one another? Without the influence of global cinematic trends and cross-cultural exchanges, we might see more pronounced differences in how colors are employed to tell stories of romance.

This thought underscores the importance of considering the broader context in which films are made. The shared visual language in these romance films may not only reflect universal emotional truths but also the interconnected nature of global cinema.

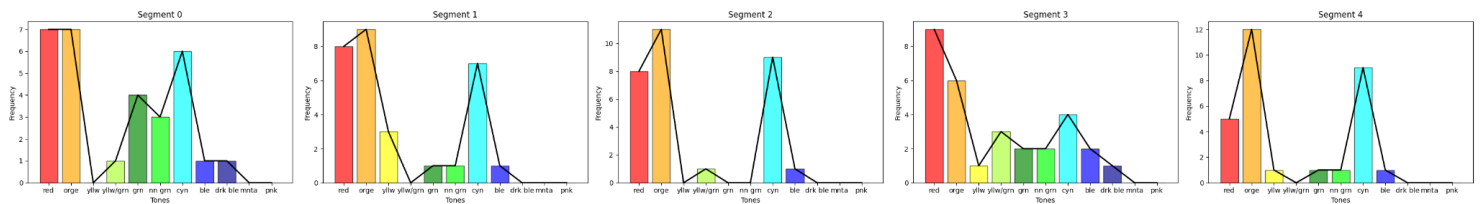
Segmented Time Series Analysis

This segmented analysis examines how the distribution of color tones varies across different parts of romance films. By splitting each movie into five segments and performing tone classification on each segment, we can observe how color usage changes throughout the length of the films.

Note: The graphs here have been condensed due to space issues. To see them more clearly, they can be found in the file `Combined_K-means_Time_Series_Histogram_Analysis.ipynb`

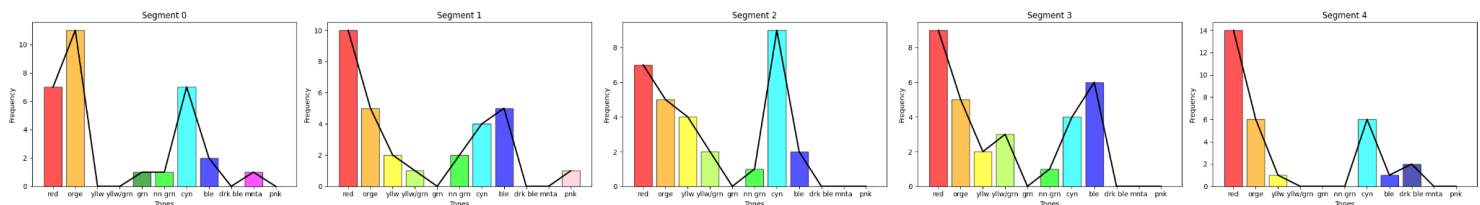
English Films

English Movies - Tone Distribution Across Segments



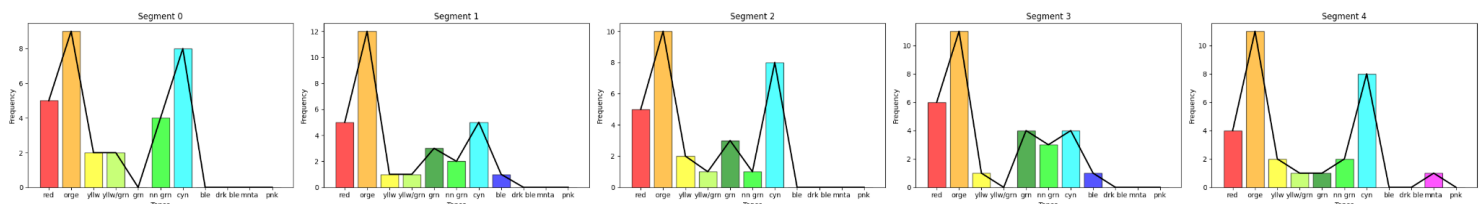
Hindi Films

Hindi Movies - Tone Distribution Across Segments



Korean Films

Korean Movies - Tone Distribution Across Segments



After analyzing the segmented time series data for English, Hindi, and Korean films, several common patterns and unique differences emerge

1. Consistency in Warm Tones:

- Across all three languages, warm tones (particularly red and orange) consistently dominate the color palette throughout the films. This indicates a shared emphasis on warmth and emotional resonance in romance films, regardless of cultural context.

2. Variation in Cool Tones:

- While cool tones like cyan and blue are present across all languages, their distribution varies.

3. Dynamic Use of Color in Hindi Films:

- Hindi films tend to show more varied use of color across segments, which seems to reflect Bollywood's vibrant and expressive visual style.

4. Stable Visual Narrative in Korean Films:

- Korean films exhibit a high degree of color consistency, with a particular focus on orange and cyan tones. This stability suggests a deliberate approach to maintaining a cohesive visual tone.

5. English Films' Balanced Approach:

- English films strike a balance between warm and cool tones, with consistent use of reds and oranges complemented by varying levels of blue and cyan. This approach reflects a balanced visual narrative that supports the story's romantic and dramatic elements.

The segmented time series analysis reveals common strategies and unique approaches in using color across different languages. While the overall use of warm tones is a unifying factor, the variation in cool tones and the dynamic or stable use of color across segments highlight the distinct visual storytelling techniques employed in English, Hindi, and Korean romance films.

Analysis of Saturation, Brightness, and Contrast

In this section, I conducted an analysis of the saturation, brightness, and contrast of the colors identified through the initial K-Means clustering. The goal was to determine if there is a bias in color characteristics based on language, which might indicate the use of specific color filters or aesthetic trends within the movies of each language. Code found in:

(Language)_K_means&_Color_Analysis.ipynb.

Note: I understand that K-means doesn't give us the exact colors used in the films as they are mean values, but they should be close enough to represent the original values.

How the Values Were Calculated:

1. Saturation and Brightness:

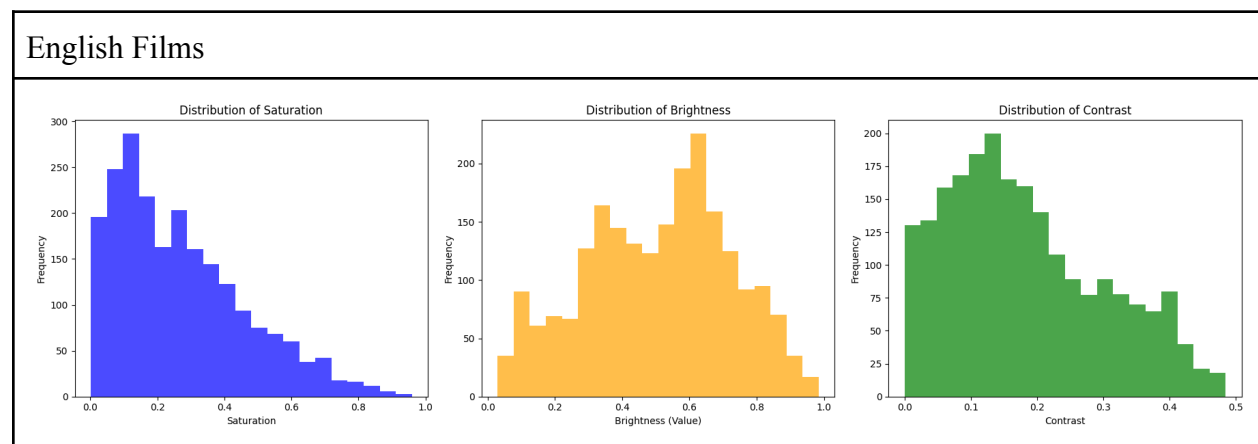
- **Conversion to HSV:** The RGB colors identified from K-Means clustering were converted into the HSV (Hue, Saturation, Value) color space.
- **Saturation:** This component measures the intensity or purity of the color, where higher values represent more vivid, pure colors, and lower values indicate more muted, grayish tones.
- **Brightness (Value):** This represents the lightness of the color, where higher values correspond to brighter colors, and lower values to darker shades.

2. Contrast:

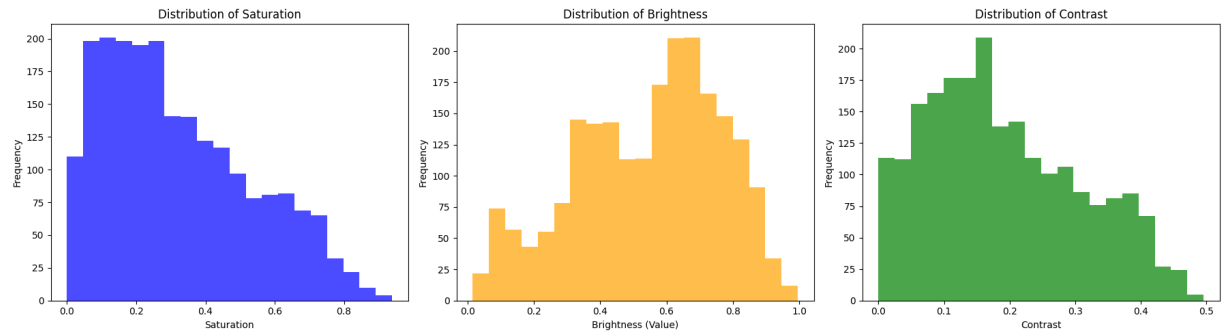
- Contrast, in the context of individual colors, can be approximated by evaluating how far the brightness (value) deviates from the mid-point on the scale (0 to 1). A high-contrast color would have a brightness value that is significantly different from 0.5.

Graphs of Saturation, Brightness, and Contrast

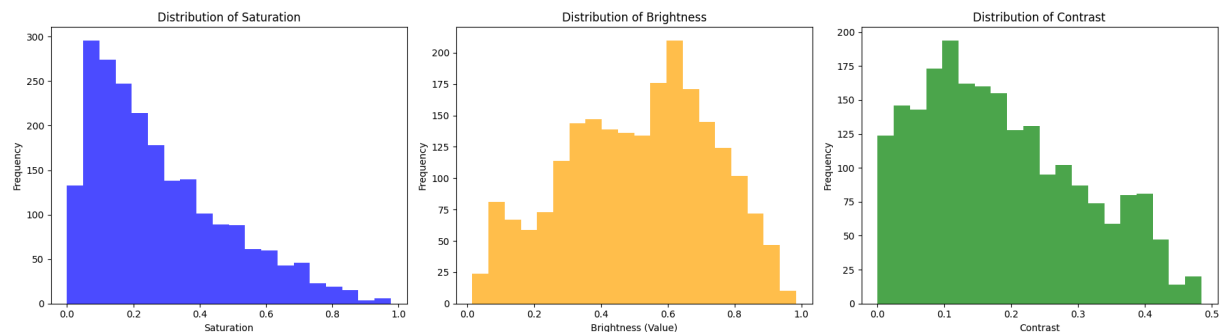
These are the distributions of values found in the K-means initial Clustering.



Hindi Films



Korean Films



1. Saturation:

- **Common Trend:** Across all three languages, the saturation distribution is skewed towards lower values, indicating a general preference for more muted colors. This suggests that filmmakers often opt for subtler, less intense tones.
- **Differences:**
 - **Hindi Films:** Among the three, Hindi films show a slightly broader range of saturation values, with a notable inclusion of more vibrant colors. This aligns with Bollywood's reputation for dynamic and expressive visuals, where more saturated colors might be used to heighten emotional intensity and visual spectacle.

2. Brightness:

- **Common Trend:** The brightness distributions across all three languages are relatively similar, with a peak around the 0.5 to 0.7 range. This suggests that filmmakers in all three contexts favor moderately bright colors, allowing for a balanced visual tone that supports both lighter and darker scenes.

- **Differences:**
 - **Hindi Films:** The distribution of brightness in Hindi films shows a wider spread, indicating a more varied use of brightness levels. This variability reflects the diverse and vibrant visual style of Bollywood.
 - **Korean Films:** Korean films maintain a balanced brightness distribution, similar to English films, but with a slightly broader spread towards both lighter and darker tones.

3. Contrast:

- **Common Trend:** The contrast distributions are remarkably similar across all three languages, with peaks around the 0.2 to 0.3 range. This indicates a preference for moderate contrast, where the visuals are neither too sharp nor too soft, supporting a balanced and versatile visual style.
- **Differences:**
 - **Korean Films:** Korean films show a slight tendency towards lower contrast values, suggesting a preference for softer visuals that contribute to a more subdued and atmospheric aesthetic.
 - **Hindi Films:** While the contrast distribution in Hindi films is similar to the others, there is a slightly higher frequency of lower contrast values, which might be used to create a more dreamlike or romantic atmosphere.

The comparative analysis of saturation, brightness, and contrast across English, Hindi, and Korean romance films reveals common strategies and distinct cultural influences in color. While all three languages share a similar approach to creating a visually balanced and emotionally resonant narrative, the subtle differences in color characteristics highlight the unique visual identities of each film industry.

Analysis of Color Temperature

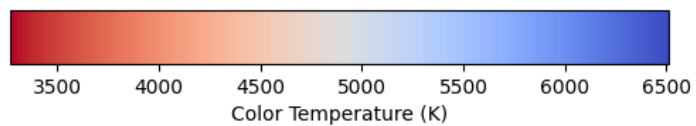
Here, I explored color temperature to understand whether certain languages tend to use warmer or cooler tones overall. This analysis helps identify any biases towards warm or cool color schemes in the films, which could indicate cultural preferences or stylistic trends within each film industry.

Temperature Calculation

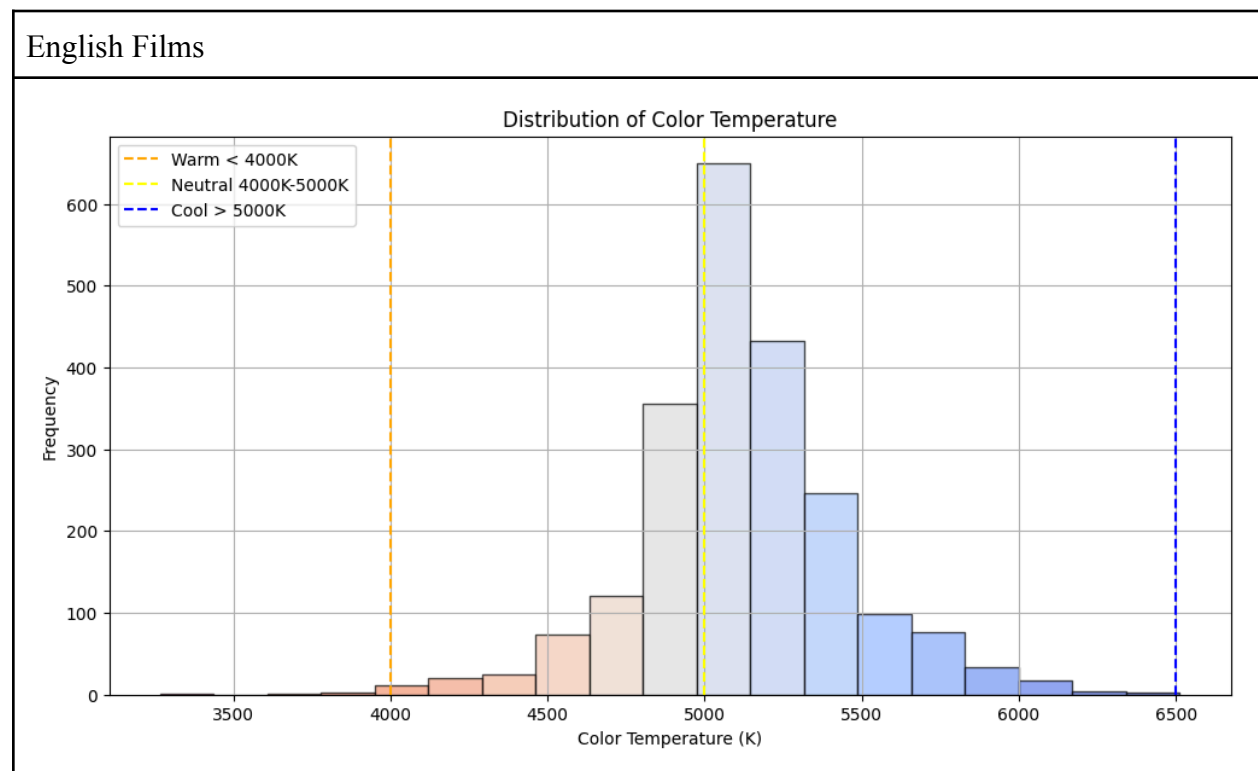
The color temperature of a color is often associated with how "warm" or "cool" it appears. Warmer colors are typically closer to red, orange, and yellow, while cooler colors lean towards blue and green.

```
def calculate_color_temperature(r, g, b):
    # Approximate color temperature calculation (based on RGB)
    # This is a simplified model; more complex models use
    chromaticity calculations
    if r > b:
        # Warmer colors
        temperature = 5000 + (r - b) * 2000 / 255
    else:
        # Cooler colors
        temperature = 5000 - (b - r) * 2000 / 255
    return temperature
```

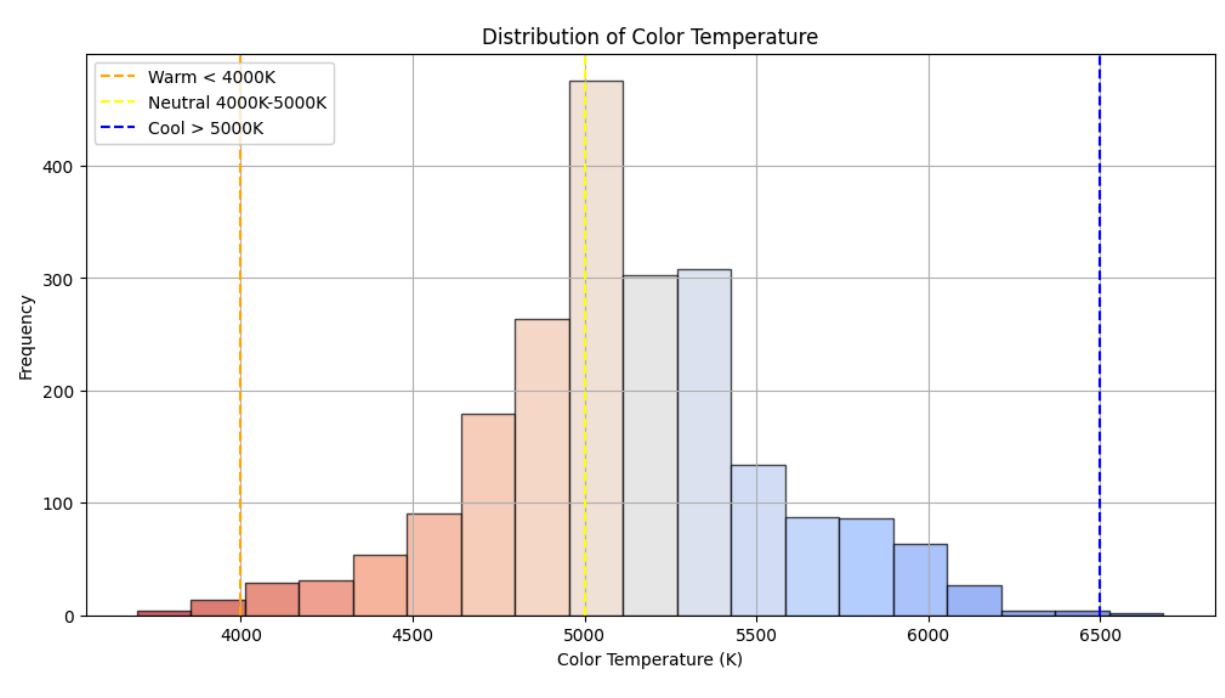
The code approximates color temperature by comparing the red (r) and blue (b) components of an RGB color. In general, colors with a higher red component compared to blue are considered "warmer," while colors with more blue are considered "cooler."



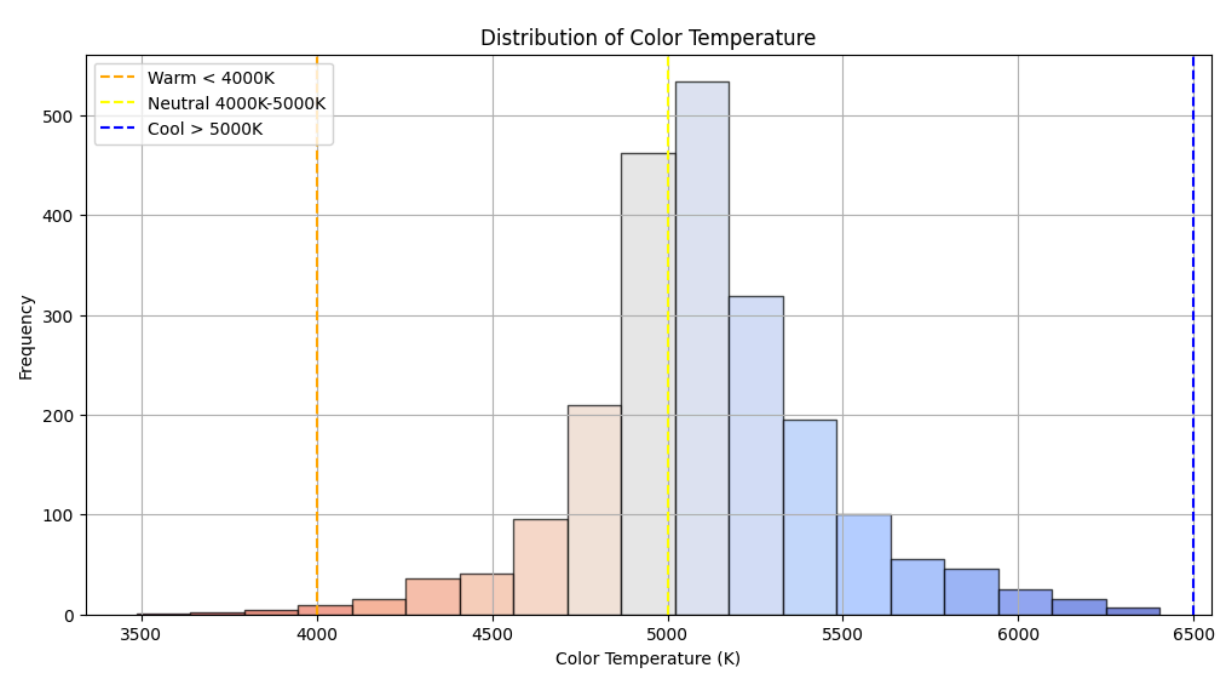
Graphs of Color Temperature



Hindi Films



Korean Films



1. Common Trend Towards Neutral Tones:

- Across all three languages, there is a noticeable peak around the neutral range (5000K). This suggests that neutral tones are a central element in romance films, likely because they offer versatility and balance, supporting a wide range of narrative contexts without overwhelming the viewer with extreme warmth or coolness.

2. Distinct Preferences:

- **English Films:** Lean towards neutral to cool tones, with a limited use of warm colors. This supports a more subdued aesthetic that aligns with English-language romance films' often dramatic or somber themes.
- **Hindi Films:** Show a broader distribution towards warmer tones. Warmer tones are known to complement darker skin tones, enhancing their natural glow and providing a visually appealing contrast. In Bollywood, where actors typically have darker skin tones, using these colors might be a deliberate choice to create a more flattering and vibrant visual aesthetic.
- **Korean Films:** Exhibit a strong preference for neutral to cool tones, with a sharp peak at 5000K and an emphasis on cooler colors. The preference for cooler tones may reflect cultural influences or cinematic trends within Korean cinema, where visual subtlety and emotional depth are often prioritized over vibrant or intense color schemes.

As we've now seen multiple times, the analysis across English, Hindi, and Korean romance films reveals both shared visual strategies and distinct cultural preferences. While neutral tones are a common thread, the varying emphasis on warm or cool colors highlights each film industry's unique aesthetic and emotional priorities.

Note: A bit more analysis was done in the notebooks comparing all of these features. If you'd like to see it, its found in these notebooks: **(Language)_K_means&_Color_Analysis.ipynb**.

Final K-Means Clustering

So, keeping in mind all the analysis I've done so far, I wanted to look at the results of applying K-Means clustering again to the set of colors I obtained from the initial K-Means run. The goal was to condense the approximately 2,600 colors into a set of 30 representative colors for each language. This final analysis will allow us to see how the visual styles of English, Hindi, and Korean romance films compare when distilled down to their core color palettes.

Final K-Means Colors

English Films

	#a75342		#904c2b		#4d3729		#a17d63		#5e4c3b
	#171512		#866e58		#6e6051		#2f2a23		#d9a65a
	#bcac90		#a5977d		#cfc4ab		#838077		#d1d3ca
	#b5b6b1		#5e9e35		#575b5a		#69706e		#dee7e6
	#8a9292		#9da5a5		#434848		#308896		#a3c0d2
	#6f9fc2		#244258		#688296		#47647f		#4248d7

Hindi Films

	#b13e39		#c1bfbe		#953d24		#a8674d		#5a3a2b
	#836753		#bc9679		#918170		#c4b09c		#665743
	#342d25		#d8c7a7		#9f988b		#171613		#d4b043
	#82a051		#434943		#6a706c		#d6dad9		#9ea8a9
	#348ea7		#63a5cb		#4b5b65		#2b4658		#607e97
	#838f9e		#9db3d1		#4654ab		#c044ba		#bf4269

Korean Films

#ba5f5c	#a85232	#0f0c0b	#7a4722	#483829
#8b765f	#766048	#5a4b39	#a78a64	#1d1b18
#302d29	#ac9f82	#c3b595	#c3a84e	#ceccb7
#8a8a7f	#849942	#636860	#d9e1db	#aeb8b4
#4e5451	#9aa19e	#40b1b8	#687779	#becdd0
#328396	#375a64	#324348	#718e9a	#8aabc0

1. Distribution of Colors:

- **English Films:** The English film palette is characterized by earthy browns, muted blues, and neutral grays, with an emphasis on more subdued tones. This reflects a preference for cooler and neutral colors, contributing to a more realistic and dramatic aesthetic.
- **Hindi Films:** The Hindi film palette is more diverse and vibrant, with a rich mix of warm tones (reds, oranges) and cool tones (blues, greens). This reflects the dynamic and expressive visual style of Bollywood, where color is used to enhance emotional intensity and visual spectacle.
- **Korean Films:** The Korean film palette strikes a balance between warm, cool, and neutral tones, with an emphasis on muted and pastel colors. This suggests a preference for softer, more subtle visuals that align with the emotionally nuanced and introspective nature of Korean cinema

The English films favor a more controlled and cooler palette, reflecting a restrained aesthetic. In contrast, Hindi films embrace a broader and more vibrant range of colors, indicative of Bollywood's dynamic and expressive style. Korean films, with their tendency towards pastel tones, offer a softer, more subdued palette that supports their introspective and emotionally rich narratives.

2. Brightness:

- **English Films:** The brightness levels are moderate, with few extremes, indicating a balanced visual tone that avoids overly bright or dark scenes. This consistency supports the realistic and somber tone of English romance films.
- **Hindi Films:** The brightness distribution is broader, with both bright and dark tones represented. This contributes to the dynamic visual contrast seen in Bollywood films, allowing for dramatic shifts in mood and tone.

- **Korean Films:** The brightness levels are also moderate, but with a slight lean towards lighter tones due to the pastel colors. This supports the films' calm and reflective atmosphere.

While English and Korean films maintain a balanced brightness, Hindi films utilize a wider range, contributing to the visual dynamism and intensity of Bollywood cinema. The tendency towards lighter, pastel tones in Korean films creates a gentle and serene aesthetic that contrasts with the more dramatic visual styles of the other two languages.

3. Saturation:

- **English Films:** The saturation is generally low to moderate, with most colors appearing muted. This aligns with the films' focus on realism and emotional depth, where colors are used subtly to support the narrative.
- **Hindi Films:** The saturation is higher, with several colors appearing vivid and intense. This use of vibrant colors aligns with Bollywood's emphasis on emotional expressiveness and visual impact.
- **Korean Films:** The saturation is low to moderate, with a noticeable lean towards pastel tones. This subdued saturation enhances the films' introspective and gentle visual style.

The muted saturation in English and Korean films contrasts with the vivid and saturated colors in Hindi films. This difference reflects the cultural and cinematic priorities of each industry, with Bollywood favoring bold and expressive visuals, while English films emphasize subtlety and realism, and Korean films explore a softer, more contemplative aesthetic.

4. Color Temperature:

- **English Films:** The palette includes a mix of neutral and cool tones, with very few warm colors.
- **Hindi Films:** The palette features a significant amount of warm tones, reflecting Bollywood's vibrant and emotionally expressive aesthetic.
- **Korean Films:** The palette includes cooler tones, with some neutral and warmer colors, and a noticeable trend towards pastel tones.

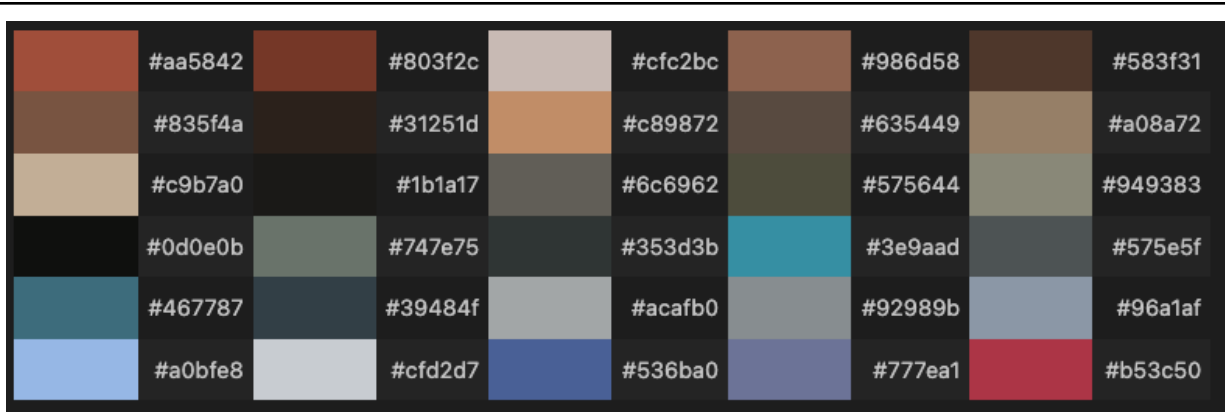
The cooler tones in English and Korean films contrast with the warmer tones in Hindi films, reflecting different cultural and aesthetic preferences. The cooler palettes in English and Korean films support a more subdued and introspective narrative. In contrast, the warmer palette in Hindi films aligns with the vibrant and expressive visual style of Bollywood. The pastel tones in Korean films add a unique softness and delicacy to the overall aesthetic.

The combined analysis of the final K-Means clustering for English, Hindi, and Korean romance films reveals how each film industry uses color to create a distinct visual language.

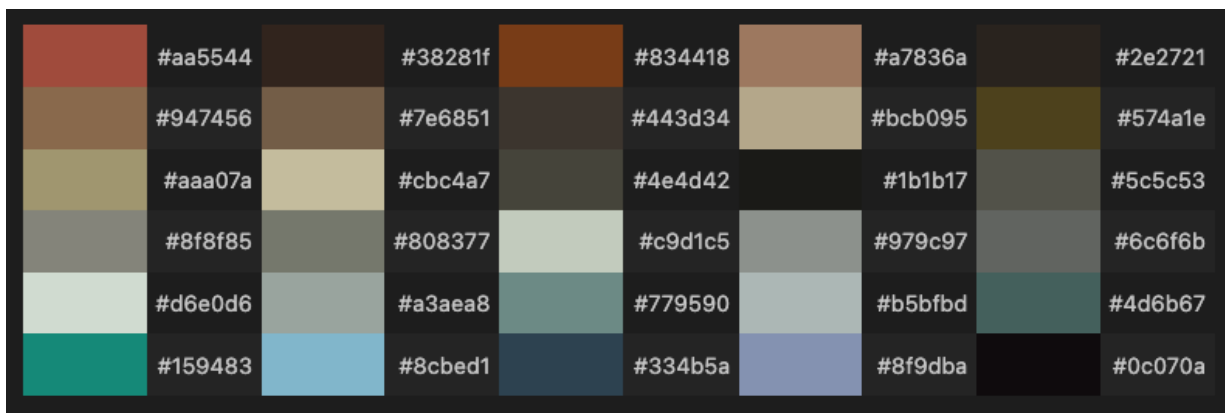
Note: I also did a final K-Means analysis on each segment, to get a set of colors representing each segment of the films'. The colors found are quite interesting and convey similar sentiments to the grouped one but in a more vivid manner, the colors can be found in the appendix and the code in these notebooks: **(language)_K_means_Segmented_Analysis.ipynb**

In this section, I explore the application of Gaussian Mixture Model (GMM) clustering to the color data from English, Hindi, and Korean romance films. The goal was to see if GMM could reveal more nuanced patterns compared to K-Means clustering, especially in cases where the data might not be well-separated. By using the Bayesian Information Criterion (BIC) to determine the optimal number of clusters, I aimed to balance the model's complexity with its fit to the data.

English Films									
	#070707		#9e5236		#825e4a		#5b3b28		#483b32
	#734c30		#a47e62		#8d6f59		#5b4e41		#34291f
	#1e1915		#f6b44e		#968872		#beb5a4		#6f6c57
	#767672		#cccab1		#5a5b56		#8d8f8a		#a6aaa1
	#424743		#97a39c		#c1dad9		#d8dcde		#435862
	#0d2539		#93b1cb		#94999d		#65809b		#342dc6



Korean Films



1. GMM Clustering for English Films:

- **Cluster Distribution:** The GMM clusters for English films show a mix of muted and neutral tones, similar to what we observed with K-Means clustering. The palette includes earthy browns, grays, and muted blues, with a few warm tones like burnt orange.
- **Observations:** The clusters do not seem significantly more distinct than those from K-Means. The colors are still predominantly subdued, with few bright or saturated tones. This suggests that, for English films, the data may not benefit much from the probabilistic approach of GMM, as the overall visual style favors muted, realistic colors.

2. GMM Clustering for Hindi Films:

- **Cluster Distribution:** The Hindi film clusters from GMM are similar in nature to the K-Means clusters, with a range of warm and cool tones. The colors are somewhat less saturated and vibrant than in the K-Means clusters, which might be due to the broader range of colors each cluster represents.
- **Observations:** While GMM might capture broader color categories, it doesn't seem to enhance the distinction between clusters significantly. The overall vibrancy of

Bollywood's visual style is still present, but the colors appear slightly more blended and less distinct, reinforcing the decision to stick with K-Means for capturing the dynamic visual aesthetics of Hindi films.

3. GMM Clustering for Korean Films:

- **Cluster Distribution:** The GMM clusters for Korean films also show a muted and balanced palette, with a noticeable tendency towards cooler tones, similar to what was observed with K-Means. There are also pastel tones, though they are less distinct and more blended than in the K-Means clusters.
- **Observations:** The pastel and cooler tones that characterize Korean films are still present, but they appear less pronounced in the GMM clusters. The clusters are more vague, and the visual subtlety that is a hallmark of Korean cinema might be better captured through the more distinct clusters provided by K-Means.

Upon comparing the GMM clusters with those from K-Means, it's clear that while GMM offers a probabilistic approach to clustering, it doesn't significantly enhance the distinctiveness of the clusters in this context. The resulting colors are somewhat more blended and less saturated, which may dilute the visual characteristics that define each language's film style. This outcome supports the decision to proceed with K-Means clustering, as it provides more distinct and representative color clusters that align better with the aesthetic and emotional tones of English, Hindi, and Korean romance films.

Note: The code + how I went about doing this can be found in the respective notebooks **(language)_GMM_Analysis.ipynb**. I also analyzed the tonality of the colors, which is similar to K-Means but with colors produced by GMM. What I found was that it was almost exactly the same, therefore I'm not including it in the report, but you can see the analysis in this notebook: **Combined_K-means_Time_Series_Histogram_Analysis.ipynb**

Linguistic Link To the Colors

What I've found so far in this project has largely confirmed common knowledge, which, while valuable, is somewhat expected. However, I had hoped to uncover something more unexpected or nuanced. At this stage, I started searching for materials that could provide a deeper connection to the project, and I remembered watching some videos about how different civilizations developed color terms in their languages and how these linguistic distinctions might affect color perception. This is still a hotly debated topic, so take it as food for thought rather than a definitive explanation.

The Videos

▶ Why The Ancient Greeks Couldn't See Blue - AsapScience

▶ All The Colours, Including Grue: How Languages See Colours Differently - Tom Scott

▶ The surprising pattern behind color names around the world - Vox

Two specific points from these videos sparked my curiosity about how language might influence the colors used in film. First, the AsapScience video mentioned that the Himba people of Namibia do not have a term for the color blue in their language, which leads them to take longer to distinguish blue from a set of greens compared to someone who has a word for blue. Second, Tom Scott made an insightful comment when defining linguistic relativism: "It's about the words we know and the languages we learn, which influence how we interpret the world." This made me wonder whether the specific languages and their interpretations of color could be influencing the themes I noticed in my K-Means clustering results. Specifically, I observed darker and duller colors in English films, brighter and more vibrant colors in Hindi films, and soft pastels in Korean films.

Investigating Etymological and Cultural Links

English and Korean Film

My initial research focused on whether there were etymological reasons behind these observed color themes. I wondered if I could find any connections between the meanings of color terms in each language and the visual aesthetics I had identified. Unfortunately, despite extensive searching (limited to English-language sources, as I am unable to research in Hindi or Korean), I didn't find any specific etymological explanations. However, I did come across an interesting paper titled "Cross-Cultural Comparison of Color Preferences between English and Korean Subjects," which provided some valuable insights.

In the study, the authors aimed to create “a cross-cultural framework that is capable of helping designers in England develop websites that are attractive for Koreans and vice versa” (Choi, 170). They found that “Korean subjects prefer soft, bright, and pastel tone colors, whereas English subjects prefer strong, dark, dull, and intense colors” (Choi, 175). This aligns perfectly with the color themes I observed in my analysis.

The paper offered further insight into why these preferences might exist: “Koreans prefer natural colors, and the result of the above experiment confirmed this. The relationship between attitudes towards nature and color schemes is also supported by recent compilations of colors. Here, 352 different kinds of colors were named based on nature-related items, such as names of animals, plants, and natural materials” (Choi, 170). Essentially, the preference for pastel colors in Korean films could be attributed to the linguistic and cultural association of color names with natural environmental elements, which tend to be more pastel-toned. “The majority of colors in the Korean results could be related to natural items and the four seasons with their changing colors” (Choi, 176). However, the paper does not delve into the reasoning behind the preference for strong, dark, and intense colors among English subjects. My best guesses are that it could be related to historical factors, such as the Industrial Revolution, or environmental factors based on the moody climate of the UK, but this doesn’t extrapolate well to the USA. However, this remains speculative.

Hindi Film

Regarding the vibrant colors found in Hindi cinema, specifically Bollywood, I found an interesting book titled “Understanding Bollywood: The Grammar of Hindi Cinema.” While the book is not specifically focused on linguistics, it offers cultural insights that help explain the use of color. The author describes Bollywood films as embracing “an aesthetic of overstatement and excess—an aesthetic in which, much like Broadway or a drag performance, outrageousness is favored over subtlety” (Ulka, 84). This aligns with the bright colors observed in the K-Means analysis.

The book further explains that “Many scholars trace Bollywood’s unique visual style to the Parsi theater, a hybrid European and Indian form of drama popular in India in the late nineteenth and early twentieth centuries. The frontal view, theatrical staging, and overall ‘structure of excess,’ in addition to a thematic focus on love, characterized the Parsi theater and can be seen as a prototype of Bollywood” (Ulka, 84). The bright colors I observed are further significant in regards to romance as the author states, “Bright colors usually signify a moral universe and the possibility of love and sexuality” (Ulka, 86). Moreover, love stories in Bollywood tend to be filled with celebrations, such as weddings and the Hindu holiday of Holi, which are “always full of color” (Ulka, 86).

Conclusion

This exploration suggests that linguistics and culture play significant roles in shaping the colors used in films, influencing the visual language of cinema in subtle yet profound ways. While the connection between language and color perception remains a complex and debated topic, it's clear that the cultural context of a language deeply impacts the aesthetic choices in film, making this an important consideration in any cross-cultural analysis of cinematic visuals.

Conclusion

This study aimed to explore how language and cultural contexts influence the use of color in romance films from English, Hindi, and Korean cinema. By analyzing the color palettes through K-Means clustering, evaluating brightness, saturation, contrast, and color temperature, and examining potential linguistic links, several key findings emerged.

Key Findings

1. **K-Means Clustering:** The final K-Means clustering revealed distinct color palettes across the three languages. English films favored earthy browns, muted blues, and neutral grays, reflecting a more subdued and realistic aesthetic. Hindi films, in contrast, embraced vibrant and diverse colors, with a rich mix of warm tones and dynamic hues that align with Bollywood's expressive visual style. Korean films balanced between warm, cool, and neutral tones, often leaning towards pastel colors, which supported the introspective and emotionally nuanced nature of Korean cinema.
2. **Brightness, Saturation, and Contrast:** The analysis of these attributes showed that Hindi films exhibited a broader range of brightness and higher saturation, reinforcing the dynamic and vibrant visuals typical of Bollywood. English and Korean films maintained moderate brightness and saturation, with English films focusing on realism and Korean films leaning towards softer, pastel tones. Contrast was moderate across all languages, contributing to a balanced visual style.
3. **Color Temperature:** English films predominantly used neutral to cool tones, contributing to their restrained and often somber aesthetic. Hindi films displayed a broader distribution towards warmer tones, which might be linked to cultural factors, such as the preference for colors that complement darker skin tones in Bollywood. Korean films favored neutral to cool tones, with a noticeable emphasis on pastel hues, reflecting a subtle and emotionally resonant visual style.
4. **GMM vs. K-Means:** While Gaussian Mixture Modelling (GMM) was explored as an alternative to K-Means, the results did not provide significantly more distinct clusters. The GMM clusters were more blended and less saturated, leading to the decision to stick with K-Means for its ability to capture more distinct and representative color clusters that align better with the visual styles of each language's films.
5. **Linguistic and Cultural Links:** The investigation into linguistic links revealed that language and culture indeed play a crucial role in shaping the colors used in films. For example, the preference for pastel tones in Korean cinema could be attributed to the linguistic association of color names with natural elements. Similarly, the vibrant colors in Hindi films reflect Bollywood's cultural emphasis on visual spectacle and emotional expression. In contrast, the darker and muted tones in English films might be influenced by historical or environmental factors, although this remains speculative.

Final Thoughts

Overall, the analysis highlights how different cultural and linguistic contexts influence the color palettes used in romance films. While there are shared visual strategies across these films—such as the universal use of warm tones to evoke emotions—the unique aesthetic choices of each film industry reflect deeper cultural and linguistic influences. This study underscores the importance of considering these factors when analyzing cinematic visuals, particularly in a genre as emotionally charged as romance.

Reflecting on the broader implications of this work, a quote from the Vox video resonates deeply: “So what does all this mean? Why does it matter? Well, it tells us that despite our many differences across cultures and societies ... there is something universal about how humans try to make sense of the world.” To extrapolate this idea further, perhaps what we’re really uncovering is how we make sense of love.

Citations

Anjaria, Ulka. Understanding Bollywood : The Grammar of Hindi Cinema, Taylor & Francis Group, 2021. ProQuest Ebook Central,
<https://ebookcentral.proquest.com/lib/bu/detail.action?docID=6451435>.

Choi, G. S., Dalke, H., Oehlmann, R., & Cottington, D. (2007). Cross-cultural comparison of color preferences between English and Korean subjects. In A. Nijholt, O. Stock, & T. Nishida (Eds.), *Social Intelligence Design 2007: Proceedings of the Sixth Workshop on Social Intelligence Design* (pp. 177-185). Universiteit Twente, Faculty of Electrical Engineering, Mathematics and Computer Science. Retrieved from
<https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=f7f7981008fd87d0ddcba6aac45123d4ae68a42f#page=177>

Appendix

List of English Movies

(Excluding animated film and black and white film)

https://www.imdb.com/search/title/?title_type=feature&release_date=1990-01-01,&num_votes=50000,&genres=romance&primary_language=en&sort=user_rating,desc

1. Forrest Gump (1994)	6. La La Land (2016)	11. The Legend of 1900 (1998)	16. Marriage Story (2019)	21. Call Me By Your Name (2017)
2. Good Will Hunting (1997)	7. Her (2013)	12. Titanic (1997)	17. Sing Street (2016)	22. About Time (2013)
3. Eternal Sunshine of the Spotless Mind (2004)	8. Groundhog Day (1993)	13. Poor Things (2023)	18. Before Midnight (2013)	23. Little Women (2019)
4. Before Sunrise (1995)	9. Big Fish (2003)	14. True Romance (1993)	19. Pride and Prejudice (2005)	24. Cast Away (2000)
5. Before Sunset (2004)	10. Cinderella Man (2005)	15. Edward Scissorhands (1990)	20. The Notebook (2004)	25. Atonement (2007)

List of Hindi Movies

(Excluding animated film and black and white film)

https://www.imdb.com/search/title/?title_type=feature&release_date=1990-01-01.&num_votes=50000.&genres=romance&primary_language=hi&sort=user_rating.desc

1. Chhichhore (2019)	6. Dilwale Dulhania Le Jayenge (1995)	11. Kal Ho Naa Ho (2003)	16. Kabhi Khushi Kabhie Gham... (2001)	21. Om Shanti Om (2007)
2. Dil Bechara (2020)	7. Andaz Apna Apna (1994)	12. Veer-Zaara (2004)	17. Ghajini (2008)	22. Jab Tak Hai Jaan (2012)
3. Zindagi Na Milegi Dobara (2011)	8. Lage Raho Munna Bhai (2006)	13. The Lunchbox (2013)	18. Rab Ne Bana Di Jodi (2008)	23. Teri Baaton Mein Aisa Uljha Jiya (2024)
4. Barfi! (2012)	9. My Name Is Khan (2010)	14. Rockstar (2011)	19. Yeh Jawaani Hai Deewani (2013)	24. Chennai Express (2013)
5. Dil Chahta Hai (2001)	10. Jab We Met (2007)	15. Kuch Kuch Hota Hai (1998)	20. Sultan (2016)	25. Murder Mubarak (2024)

List of Korean Movies

(Excluding animated film and black and white film)

<https://www.imdb.com/list/ls050921432/>

I also used this for movies to fill in for movies i couldn't find digitally:

<https://www.marieclaire.com/culture/a32253308/korean-romantic-movies/>

1. My Sassy Girl (2001)	6. Il Mare (2000)	11. Christmas in August (1998)	16. Architecture 101 (2012)	21. The Handmaiden (2016) (Not on the IMDB list but well-rated)
2. 3-Iron (2004)	7. Windstruck (2004)	12. Be With You (2018)	17. Tune in for Love (2019)	22. Spellbound (2023)
3. A Moment to Remember (2004)	8. I'm a Cyborg, But That's OK (2006)	13. 20th Century Girl (2022)	18. Marriage is a Crazy Thing (2002)	23. Decision To Leave (2022) (Not on the IMDB list but well-rated)
4. Oasis (2002)	9. Always (2011)	14. A Werewolf Boy (2012)	19. Past Lives (2023) (Not on the list IMDB but well-rated)	24. Love 911 (2012)
5. The Classic (2003)	10. Failan (2001)	15. The Beauty Inside (2015)	20. On Your Wedding Day (2018)	25. Mood of The Day (2016)

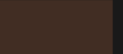
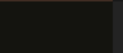
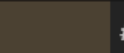
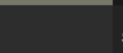
English Segmented K-Means Colors

30 colors per segment of English films.

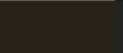
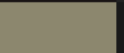
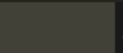
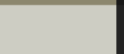
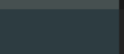
Basically each film was split into 5 segments, in which I grouped all the colors in each segment respectively and ran K-Means on those colors to get a set representing the colors found in each segment.

Code found in: **(language)_K_means_Segmented_Analysis.ipynb**

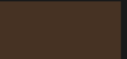
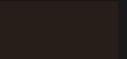
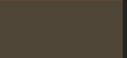
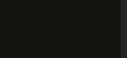
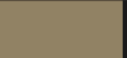
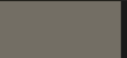
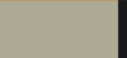
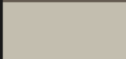
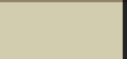
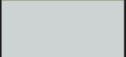
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	#b5a183		#d3c7b4		#bab3a1		#878378		#919591
	#469d48		#c0c8c2		#a5aba7		#333835		#585d5c
	#cedddb		#2fbba9		#196671		#677478		#92bfd0
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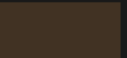
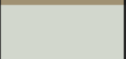
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	#b8a48a		#b49761		#c7c0ae		#9b937b		#4d4b42
	#7d7c78		#6b6b64		#a7a8a1		#d5d6cb		#7e865a
	#949c97		#348f87		#545c5f		#cedae1		#a8b1b6
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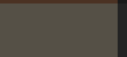
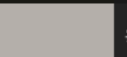
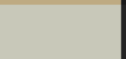
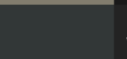
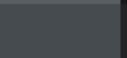
Combined Segment 2

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	#6f685c		#3c372f		#9f8f72		#827d71		#bab3a2
	#cbc7bb		#a3a08e		#dad6b8		#6cb82d		#868e8e
	#378d93		#d3dadb		#b2c0c2		#57abc6		#566064
	#3d484f		#97a0a6		#687b8a		#7da0c5		#a36f70

Combined Segment 3

	#771e10		#a65d4a		#9d4a28		#7f604b		#4d392b
	#332921		#614b37		#171310		#cd9b5a		#97856d
	#bdae98		#ad9c81		#85775e		#cac5ab		#f0f4e6
	#61655d		#d8dcd7		#c7cec5		#4b4e4c		#8b908c
	#9ca4a2		#727a7a		#aabcc1		#3a707f		#28424f
	#77a4c7		#6582a0		#4d55d0		#1a1286		#d12026

Combined Segment 4

	#a74c40		#a0624c		#954a28		#573d2b		#1e1a16
	#ad8c69		#322b24		#82725e		#645c51		#bdbab5
	#534b3f		#c9b390		#a29b90		#141210		#8f8879
	#f2c632		#d1d1c4		#66a816		#e0e8e3		#3e4140
	#9fa8a8		#8a9494		#656e6f		#afd4de		#90b1c0
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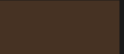
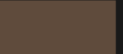
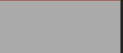
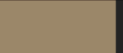
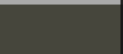
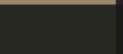
Hindi Segmented K-Means Colors

30 colors per segment of Hindi films.

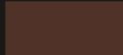
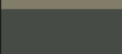
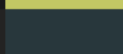
Basically each film was split into 5 segments, in which I grouped all the colors in each segment respectively and ran K-Means on those colors to get a set representing the colors found in each segment.

Code found in: **(language)_K_means_Segmented_Analysis.ipynb**

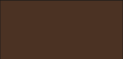
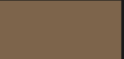
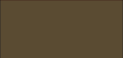
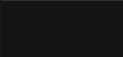
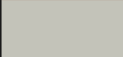
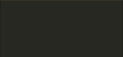
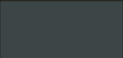
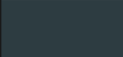
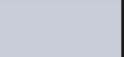
Combined Segment 0

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	#d2c5b6		#b7b5b3		#84735a		#a69478		#d4ae6a
	#8c8472		#525048		#f2c52f		#302f28		#171715
	#a1a197		#1c9d59		#d3d9d7		#5f6c6c		#328ca5
	#74abbc		#2f4553		#3d576e		#818e9a		#9db2c5
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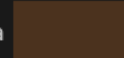
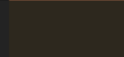
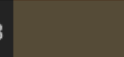
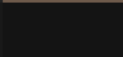
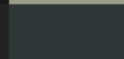
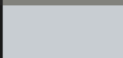
Combined Segment 1

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	#bb9476		#c7b09b		#735d47		#362f27		#1a1815
	#8f8a76		#c6c5ba		#b3ad45		#cdcf72		#a1a38f
	#525851		#d3d8d7		#687070		#2e4344		#369aa9
	#628197		#448abf		#96a6b6		#85919e		#3c4f6b
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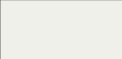
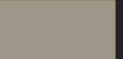
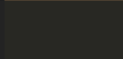
Combined Segment 2

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	#191814		#ccccd2		#2e3027		#5e6258		#789c4e
	#9fa29d		#777e76		#485150		#65b6cf		#3991ad
	#35464d		#606b70		#5b8296		#426c8d		#d1d7dd
	#355068		#80909f		#9eb3ca		#487cbe		#c1235f

Combined Segment 3

	#b2423e		#872d21		#5d4335		#97654a		#593925
	#a98e77		#796858		#c59b75		#362e23		#645643
	#c7b8a3		#181715		#c6ab3e		#a5a99a		#93a85e
	#578339		#8d918b		#6e766d		#374241		#525f61
	#398ba3		#d0d6d8		#62b2d5		#38495b		#2d5686
	#a4b3c7		#6282b4		#8894a9		#64718d		#967575

Combined Segment 4

	#d71a1a		#b03d3a		#85504c		#854025		#d9d7d6
	#a55e43		#997b6a		#462c1a		#9e9188		#d0b49e
	#564130		#71553e		#c8976b		#716758		#171513
	#f3f2ef		#aaa395		#2e2d27		#dbc826		#808777
	#4f5658		#397ba5		#6aa4cd		#9dadab9		#2e4558
	#5e7081		#8290a1		#bebebe		#6636ab		#c3437b

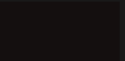
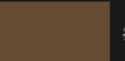
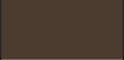
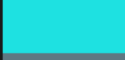
Korean Segmented K-Means Colors

30 colors per segment of Korean films.

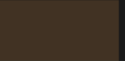
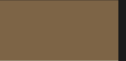
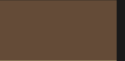
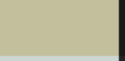
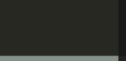
Basically each film was split into 5 segments, in which I grouped all the colors in each segment respectively and ran K-Means on those colors to get a set representing the colors found in each segment.

Code found in: **(language)_K_means_Segmented_Analysis.ipynb**

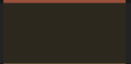
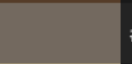
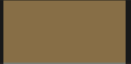
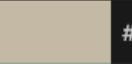
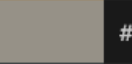
Combined Segment 0

	#91201a		#ae4d35		#13110f		#a06f48		#73573e
	#3d352c		#56493a		#8d7c66		#8d877e		#22201d
	#b8965d		#beb094		#a59c86		#bfbeab		#67685e
	#d0d6bf		#66bc5a		#d9e0d8		#6c7776		#a3adad
	#24e8e5		#2dc4c3		#3b8788		#74a7b1		#334e55
	#6a878f		#4f575a		#bccbd2		#2a7ea4		#8b98a0

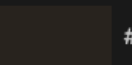
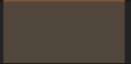
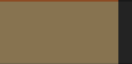
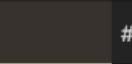
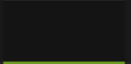
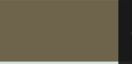
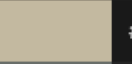
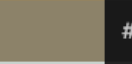
Combined Segment 1

	#ad4535		#a86347		#4d3a27		#8a6f53		#6f5940
	#1b1915		#9c8764		#4c483d		#b7af9d		#afa27c
	#d2bc5d		#b29e35		#cec8a6		#302f2c		#89887c
	#6e7957		#c1cac0		#d7e0d9		#959d98		#545e58
	#6c7471		#4dc7c4		#9ab1b2		#53959c		#32838e
	#728e96		#335461		#697bbd		#b76275		#e64a5d

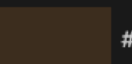
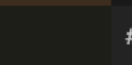
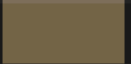
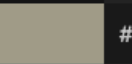
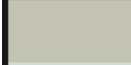
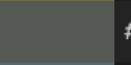
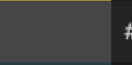
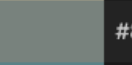
Combined Segment 2

	#c04022		#a95c49		#61452e		#a56223		#4b3e32
	#745d44		#362e26		#82786a		#1a1816		#a59378
	#aa8f61		#927b50		#c4b78e		#cbc5b2		#a19e94
	#65675c		#858980		#7cac58		#a4b1a8		#4d524f
	#ccd8d1		#56b2a6		#e1e3e4		#597073		#3b859d
	#2f434c		#788f99		#2c576d		#8ea3b3		#a6bcce

Combined Segment 3

	#b54134		#8d6148		#9b552c		#2e2823		#533b2a
	#b5a089		#5b5044		#967e5f		#403b34		#b69559
	#d0b062		#191815		#7b7359		#cac3ac		#9a9176
	#66655d		#7ca828		#dfe7e0		#747976		#d0dad3
	#b0bcb6		#9ba39f		#848e8c		#7fa5a4		#388898
	#2e444a		#435a61		#648693		#85a8d4		#bb5353

Combined Segment 4

	#a74d37		#a16455		#b6805c		#493726		#69543e
	#534739		#897c69		#a39378		#26231e		#171513
	#cab690		#7e7050		#b0851f		#e3be3e		#aaa895
	#cdcebf		#31322f		#646863		#515452		#858d8a
	#d8e1de		#76c0d0		#64777c		#32464b		#688d96
	#455d63		#95a4a8		#b0b8bb		#3184a3		#bc4ec4